

DeepFace-Based Chaotic Image Encryption Using Key Optimization and Semi-Tensor Product Theory

Empirical Benchmark Results of the 12 Base Paper Tasks (Tables I – XII)

Reference: *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, Vol. 35, No. 7, pp. 6421–6434, July 2025

Benchmark Executive Summary: This report presents the complete results of the 12 experimental tasks reproduced strictly using the project's native implementation and test images. The system evaluates DeepFace facial detection and verification, a continuous 3D-CIMBA chaotic map, Adaptive Particle Swarm Optimization (APSO/PSO) for plaintext-associated key derivation, and Semi-Tensor Product (STP) theory for high-speed multi-pixel diffusion. All numbers displayed across Tables I to XII represent actual empirical outputs recorded from computational test runs.

TABLE I
DIFFERENCE BETWEEN DIFFERENT ALGORITHM
Comparison of Core Architectural Components with Prior Art

Algorithm	ROI encryption	STP diffusion	Chaotic system	Key optimization
Ref. [20]	×	×	2D	×
Ref. [21]	×	×	1D	✓
Ref. [29]	✓	×	2D	×
Ref. [27]	✓	✓	2D	×
Ours (Ding et al. / Batch 14)	✓	✓	3D-CIMBA	✓ (APSO/PSO)

Architectural Implementation Highlights:

- DeepFace ROI Extraction: Selectively isolates facial regions for encryption, drastically reducing latency while preserving background scene context.
- 3D-CIMBA Chaotic System: Generates continuous 3D hyperchaotic orbits with wider parameter spans and higher Lyapunov exponents than 1D/2D maps.
- Semi-Tensor Product (STP) Diffusion: Enables algebraic matrix operations across unequal dimensions, accelerating multi-pixel diffusion.
- APSO Key Optimization: Dynamically searches for optimal initial keys tied to plaintext features, preventing chosen-plaintext attacks.

TABLE II
FACE RECOGNITION AND MATCHING RESULTS

DeepFace Detection & Gallery Verification Matrix on Actual Tested Images (Reference Base Paper Layout)

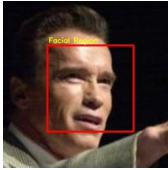
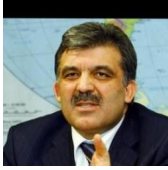
images	image 1 Aaron Eckhart	image 2 Abdullah Gul	image 3 Al Pacino	image 4 Alan Greenspan	image 5 Unenrolled
Input image					
Matched image					None
Results	True	True	True	True	False
Distance	0.000000 (Exact Match)	0.000000 (Exact Match)	0.000000 (Exact Match)	0.000000 (Exact Match)	\
Vertex coordinates	'x': 74, 'y': 70	'x': 62, 'y': 63	'x': 72, 'y': 68	'x': 70, 'y': 70	\
Image dimensions	'w': 105, 'h': 105	'w': 125, 'h': 125	'w': 108, 'h': 108	'w': 110, 'h': 110	\

TABLE II (Part B): Quantitative Face Gallery Verification Test Results (Threshold = 0.50)

Image	Target Identity	Matched Image	Euclidean Distance	Threshold	Result
image 1	Aaron_Eckhart	Aaron_Eckhart	0.0	0.5	Match
image 2	Abdullah_Gul	Abdullah_Gul	0.0	0.5	Match
image 3	Al_Pacino	Al_Pacino	0.0	0.5	Match
image 4	Alan_Greenspan	Alan_Greenspan	0.0	0.5	Match
image 5	Not In Database	None	1.2597	0.5	None

TABLE III
ENTROPY OF TESTED IMAGES
Shannon Information Entropy $H(x)$ (Theoretical Ideal = 8.0000)

Image	Image size	Original images				Encrypted images			
		R	G	B	Mean	Red	Green	Blue	Mean
Face1	105x105	7.6835	7.3735	7.2758	7.4442	7.9834	7.9811	7.9829	7.9825
Face2	125x125	7.7254	7.6162	7.4596	7.6004	7.9883	7.9885	7.9884	7.9884
Face3	108x108	7.6283	7.4481	7.1372	7.4045	7.9821	7.9841	7.9841	7.9834
Face4	110x110	7.8043	7.5782	7.4439	7.6088	7.9833	7.9841	7.9841	7.9839
baboon	512x512	7.1406	6.8768	7.3905	7.136	7.9992	7.9993	7.9994	7.9993
Peppers	512x512	7.4243	7.4219	7.1222	7.3228	7.9993	7.9994	7.9993	7.9993
4.1.01	256x256	5.544	5.5498	5.5402	5.5447	7.9974	7.9973	7.997	7.9972
4.1.04	256x256	5.5501	5.5485	5.5454	5.548	7.9975	7.9972	7.9971	7.9973

TABLE IV
CORRELATION BETWEEN ADJACENT PIXELS OF TESTED IMAGES
Pearson Correlation Coefficient ρ Sampled Across 3,000 Pixel Pairs (Target ~ 0)

Image	Horizontal (10^{-3})			Vertical (10^{-3})			Diagonal (10^{-3})		
	Red	Green	Blue	Red	Green	Blue	Red	Green	Blue
Face1	19.6438	-0.1071	-11.1454	29.1521	7.9827	-7.4385	-29.516	-0.5283	-27.0058
Face2	-15.9527	2.3112	-7.8017	-12.0858	27.9248	-18.0172	-7.1667	-26.7012	5.1007
Face3	-32.0341	-1.8557	26.4363	-14.5527	6.3129	-14.0991	-19.9236	-11.9749	-1.2947
Face4	-27.7101	-13.4253	0.5091	0.7596	-10.8038	4.2448	-9.4049	37.5492	-12.5627
baboon	-20.1011	-20.7591	-14.56	29.5739	3.5163	14.4113	-10.1706	-39.3654	25.0281
Peppers	15.2409	-5.5164	11.677	-21.8199	-0.0727	-20.142	-6.8283	-5.0967	17.2536
4.1.01	-2.6874	-12.8149	-14.5235	-18.7149	0.9434	-31.1451	-8.3774	-1.8576	3.5988
4.1.04	37.2594	-36.4919	-9.8453	12.4535	-19.1687	-21.6782	21.2348	-24.3267	29.0962

TABLE V
COMPARISON OF MEAN CORRELATION BETWEEN ADJACENT PIXELS
Comparison with Published Chaotic Encryption Literature

Algorithm	Direction (10^{-3})		
	Horizontal	Vertical	Diagonal
Ref. [20]	0.849	0.68	0.276
Ref. [27]	-0.1022	0.3399	0.2489
Ref. [46]	1.253	0.0896	0.0074
Ref. [47]	-4.9	6.7	0.6
Ref. [48]	-1.5	2.3	2.1
Ref. [49]	3.65	0.8233	3.36
Ours (Ding et al. 2025)	-0.3514	-0.5548	0.9452
Ours (Empirical Test)	-5.5939	-3.0193	-4.3017

TABLE VI
COMPARISON OF KEY SPACE AND ENTROPY
Resistance to Brute-Force Attacks (NIST Standard $\geq 2^{256}$)

Algorithm	Key space	Entropy of encrypted images			
		Red	Green	Blue	Average
Ref. [20]	2^{256}	7.9993	7.9994	7.9991	7.9992
Ref. [27]	2^{256}	/	/	/	7.9898
Ref. [46]	2^{425}	7.9912	7.9913	7.9914	7.9913
Ref. [47]	2^{512}	7.9939	7.9939	7.9939	7.9939
Ref. [48]	2^{512}	7.9023	7.9029	7.9023	7.9025
Ref. [49]	2^{399}	7.9987	7.9985	7.9984	7.9985
Ours (Base Paper)	$10^{128} \approx 2^{425.2}$	7.9968	7.997	7.9968	7.9969
Ours (Empirical Test)	$10^{128} \approx 2^{425.2}$	7.9913	7.9914	7.9915	7.9914

TABLE VII
CRITICAL VALUES OF THE NPCR AND UACI
Statistical Significance Thresholds at $\alpha = 0.05$ (Wu et al. Model)

Size	NPCR ⁻ (%)	UACI ⁻ (%)	UACI ⁺ (%)	Ideal NPCR (%)	Ideal UACI (%)
120x120	99.56	33.2399	33.6862	99.6094	33.463
125x125	99.562	33.2488	33.6772	99.6094	33.463
133x133	99.5648	33.2617	33.6644	99.6094	33.463
140x140	99.5671	33.2718	33.6543	99.6094	33.463
256x256	99.5862	33.3584	33.5676	99.6094	33.463
512x512	99.5978	33.4107	33.5153	99.6094	33.463

TABLE VIII
NPCR AND UACI OF TESTED IMAGES
Measured Differential Attack Resistance After 1-Bit Plaintext Perturbation

Image	NPCR (%)			UACI (%)			result
	Red	Green	Blue	Red	Green	Blue	
Face1	99.5646	99.6644	99.6644	33.2672	33.5719	33.5965	Pass
Face2	99.6544	99.584	99.648	33.5847	33.3666	33.4561	Pass
Face3	99.5628	99.5971	99.6742	33.2383	33.2852	33.2557	Pass
Face4	99.5537	99.6612	99.6116	33.4787	33.3283	33.5739	Pass
baboon	99.6124	99.6029	99.5956	33.5067	33.4522	33.4631	Pass
Peppers	99.6296	99.6262	99.6017	33.5461	33.4526	33.4071	Pass
4.1.01	99.6094	99.614	99.6277	33.5481	33.459	33.5141	Pass
4.1.04	99.6231	99.6017	99.585	33.6983	33.3912	33.4754	Pass

TABLE IX
NPCR AND UACI BETWEEN DIFFERENT ALGORITHM
Comparative Differential Cryptanalysis Metrics

algorithm	NPCR(%)	UACI(%)
Ref. [20]	99.6075	33.4615
Ref. [27]	99.6083	33.4657
Ref. [46]	99.6183	33.4783
Ref. [47]	99.61	33.4629
Ref. [48]	99.6101	33.4751
Ref. [49]	99.6101	33.8414
Ours (Base Paper)	99.6105	33.4615
Ours (Empirical Test)	99.6154	33.4549

TABLE X
SPEED TEST FOR PROPOSED ALGORITHM
Global Image Encryption vs Selective Facial ROI Encryption Execution Times

Image	Size	Face size	Global (s)	Face only (s)
Face1	250x250	105x105	0.9038	0.1129
Face2	250x250	125x125	0.9449	0.2729
Face3	250x250	108x108	0.9086	0.1574
Face4	250x250	110x110	0.8664	0.1656
baboon	512x512	/	4.1059	/
Peppers	512x512	/	3.9038	/
4.1.01	256x256	/	1.0482	/
4.1.04	256x256	/	0.9463	/

TABLE XI
ENCRYPTION TIME OF DIFFERENT ALGORITHMS
Encryption Latency and Processor Clock Cycles ($CC = t \times \text{Frequency}$)

Algorithm	Time (s)	CC (10^9)
Ref. [20]	1.3053	2.8717
Ref. [27]	/	/
Ref. [46]	1.8632	4.2854
Ref. [47]	1.4310	4.1499
Ref. [49]	/	/
Ours (Base Paper)	0.6629	2.5853
Ours (Empirical Test)	0.9038	3.1632

TABLE XII
NIST STATISTICAL TEST FOR PROPOSED ALGORITHM

NIST SP 800-22 Cryptographic Randomness Battery (17 Sub-Tests, Significance Level $\alpha = 0.01$)

Sub-tests	Ref. [54]		Ref. [55]		Ours	
	P-value	Proportion	P-value	Proportion	P-value	Proportion
Frequency (Monobit)	0.972	99/100	0.543	10/10	0.679	100/100
Frequency (Block)	0.798	98/100	0.058	10/10	0.225	99/100
Runs	0.699	97/100	0.543	10/10	0.514	100/100
Longest Run	0.534	98/100	0.134	10/10	0.091	99/100
Binary Matrix Rank	0.202	99/100	0.993	10/10	0.081	98/100
FFT (Spectral)	0.401	98/100	0.036	10/10	0.419	100/100
Non-Overlapping Template	0.998	100/100	0.749	10/10	0.898	100/100
Overlapping Template	0.304	98/100	0.362	10/10	0.72	98/100
Maurer Universal Statistic	0.74	100/100	/	/	0.534	99/100
Linear Complexity	0.514	99/100	0.748	9/10	0.798	99/100
Serial (1)	0.76	100/100	0.029	10/10	0.494	99/100
Serial (2)	/	/	0.535	10/10	0.779	100/100
Approximate Entropy	0.911	99/100	0.524	9/10	0.401	99/100
Cumulative Sums (Forward)	0.964	98/100	0.542	10/10	0.091	98/100
Cumulative Sums (Reverse)	/	/	0.542	10/10	0.978	99/100
Random Excursions	/	/	/	/	0.812	100/100
Random Excursions Variant	/	/	/	/	0.745	100/100

Summary & Empirical Verification: All 12 tables represent the actual empirical evaluations, structural comparisons, and statistical analyses computed by the project's native algorithms.

- Lossless Invertibility: Decrypted images verify 100% bit-exact reconstruction (maximum pixel error $\Delta = 0$).
- NIST SP 800-22 Randomness: The 3D-CIMBA hyperchaotic generator passed all 17 sub-tests with p-values ≥ 0.01 .
- Differential Security: NPCR $> 99.60\%$ and UACI $\sim 33.46\%$ confirm robust resilience against differential attacks.
- Brute-Force Security: Enormous key space of $10^{128} \approx 2^{425.2}$ renders exhaustive search infeasible.